

Simple Command Sequencer Generator

Autocoder User Guide

1 Description

The purpose of this generator is to provide a user-friendly way of defining named command sequences for the Simple Command Sequencer component. The generator takes a YAML model file as input which declares a suite of sequences – each a named, ordered list of sub-commands (and optional sleeps) – and autocodes four things from it: the Ada sequence step tables, bundled with the suite’s frame-pool size into a `Config` constant consumed by the sequencer’s `Init` routine; a set of first-class per-sequence wrapper commands so that each sequence appears on the sequencer instance as its own typed, operator-callable command; an operator-side command builder package for those wrapper commands; and the record type of the instance’s periodic summary packet.

Note the example shown in this documentation is used in the unit test of this component so that the reader of this document can see it being used in context. Please refer to the unit test code for more details on how this generator can be used.

2 Schema

The following pykwalify schema is used to validate the input YAML model. Model files must be named in the form *optional_name.assembly_name.command_sequences.yaml*, where *optional_name* is the specific name of this set of sequences (only needed if more than one suite is defined in the assembly) and *assembly_name* is the assembly the sequences will be used in. Generally this file is created in the same directory or near to the assembly model file. The schema is commented to show each available YAML key and what it accomplishes; even without knowing the specifics of pykwalify schemas, you should be able to glean the available structure from the file below.

```
1  # Schema for command_sequences YAML files
2  # This defines the structure and validation rules
3
4  type: map
5  mapping:
6    name:
7      required: true
8      type: str
9      desc: |
10         The name of the command sequences package. This will be used as the
11         Ada package name, so it must follow Ada naming conventions.
12
13    description:
14      required: false
15      type: str
16      desc: |
17         Description of this command sequences package. This will appear in
18         the generated Ada code comments.
19
20    preamble:
21      required: false
22      type: str
```

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23     desc: |
24         Custom preamble text to include in the generated Ada package.
25
26 num_concurrent_sequences:
27     required: true
28     type: int
29     range: { min: 1 }
30     desc: |
31         The number of sequence frames (concurrently running sequences) a
32         sequencer instance initialized with this suite allocates. This also
33         sizes the suite's generated summary packet ground type (one
34         Sequence_Frame_Summary per frame), so the frame pool and the packet
35         layout can never disagree. The upper bound -- how many frame
36         summaries fit in one packet buffer -- is enforced by the model
37         against the project configuration.
38
39 command_timeout_seconds:
40     required: false
41     type: int
42     range: { min: 0, max: 2147483 }
43     desc: |
44         Default timeout, in whole seconds, that the sequencer will wait for
45         a sub-command's response before declaring the sequence failed. Used
46         for any sequence in this suite that does not specify its own
47         command_timeout_seconds. Bounded so the millisecond count always
48         fits an Ada.Real_Time.Time_Span.
49
50 with:
51     required: false
52     type: seq
53     desc: |
54         List of additional Ada packages to "with" in the generated sequences
55         package. Use this when a step's arg expression references a package
56         that is not pulled in automatically, e.g. an enum package for a
57         qualified enumeration literal.
58     sequence:
59         - type: str
60
61 sequences:
62     required: true
63     type: seq
64     desc: |
65         List of command sequence definitions. Each sequence defines a named
66         command that executes a series of sub-commands.
67     sequence:
68         - type: map
69           mapping:
70             name:
71                 required: true
72                 type: str
73                 desc: |
74                     Name of this sequence. This will become a command that can be
75                     executed. Must follow Ada naming conventions.
76
77             description:
78                 required: false
79                 type: str
80                 desc: |
81                     Description of what this sequence does. Will appear in generated
82                     code and documentation.
83

```

```

84     arg_type:
85         required: false
86         type: str
87         desc: |
88             The Ada type for this sequence's argument. If not specified,
89             the sequence takes no arguments. Must be a valid Ada type name,
90             possibly qualified (e.g., "Power_Mode.T" or "Packed_U16.T").
91
92     wait_for_command_completion:
93         required: false
94         type: bool
95         default: true
96         desc: |
97             Default setting for whether to wait for each sub-command to
98             complete before executing the next one. Individual steps can
99             override this.
100
101     continue_on_failure:
102         required: false
103         type: bool
104         default: false
105         desc: |
106             If true, the sequence will continue executing even if a
107             ↪ sub-command
108             fails. If false, the sequence will abort on the first failure.
109
110     response_behavior:
111         required: false
112         type: str
113         enum: ['send_after_sequence_start',
114             ↪ 'send_after_sequence_completion']
115         desc: |
116             When the sequencer replies to this sequence's own command:
117             send_after_sequence_start (the default) acknowledges as soon as
118             the sequence is started; send_after_sequence_completion defers
119             the reply until the sequence finishes, carrying its final
120             success or failure. Static per-sequence configuration; the
121             generic Run_Sequence command uses it too.
122
123     command_timeout_seconds:
124         required: false
125         type: int
126         range: { min: 0, max: 2147483 }
127         desc: |
128             Override the suite-level default timeout for sub-command
129             responses, in whole seconds, for this sequence only. Bounded
130             so the millisecond count always fits an
131             Ada.Real_Time.Time_Span.
132
133     sequence:
134         required: true
135         type: seq
136         desc: |
137             The ordered list of steps to execute. Each step is either a
138             command dispatch (`command:` + optional `arg:`) or an in-sequence
139             sleep (`sleep_ms:` + integer milliseconds). The two forms are
140             mutually exclusive; exactly one must be present per step.
141             Validation is enforced in the model loader.
142         sequence:
143             - type: map
144               mapping:

```

```

143     command:
144         required: false
145         type: str
146         desc: |
147             The command to execute in the format
148             ↪ "Component.Command_Name".
149             The component must be a valid component instance name in
150             ↪ your
151             assembly, and the command must exist on that component.
152             Mutually exclusive with `sleep:`.
153
154     arg:
155         required: false
156         type: str
157         desc: |
158             An Ada expression that evaluates to the argument for this
159             command. You can reference the parent sequence's argument
160             using "Arg.field_name" syntax. If the command takes no
161             argument, omit this field. Only valid alongside
162             ↪ `command:`.
163
164     Examples:
165     - "Arg" - pass through the entire sequence argument
166     - "Arg.Power_Mode" - pass a field from the sequence
167       ↪ argument
168     - "(Mode => Arg.Mode, Enable => True)" - construct a
169       ↪ record
170     - "42" - a literal value
171
172     wait_for_completion:
173         required: false
174         type: bool
175         desc: |
176             Whether to wait for this specific command to complete
177             ↪ before
178             proceeding. If not specified, uses the parent sequence's
179             wait_for_command_completion setting. Only valid alongside
180             `command:`.
181
182     sleep_ms:
183         required: false
184         type: text
185         desc: |
186             Milliseconds to pause the sequence. Not bus-routed; the
187             sequencer frame parks itself for the duration without
188             issuing any command. Mutually exclusive with
189             `command:`/`arg:`/`wait_for_completion:`.
190
191             Either a static integer (0 .. 2**31 - 1, so it always fits
192             an Ada.Real_Time.Time_Span; enforced by the model), or a
193             dynamic reference into the sequence's argument using the
194             same "Arg[.field]..." syntax as dynamic command args. A
195             dynamic sleep's referenced field must be a
196             Packed_Natural.T, which the resolver validates at
197             execution time.
198
199     Examples:
200     - sleep_ms: 3000 # sleep 3 seconds
201     - sleep_ms: Arg.Settle_Time # duration from the sequence
202       ↪ argument

```

```

197 # Additional validation rules (enforced in the model Python code):
198 # - Command format must be "Component.Command" with valid identifiers, or
199 #   the reserved name "Sleep"
200 # - Referenced types in arg_type must exist in the system
201 # - Argument expressions must be valid Ada syntax
202 # - Component and command names must match instances in the assembly

```

3 Example Input

The following is an example command sequences input YAML file. It defines eleven sequences exercising the main features: statically-armed commands and static sleeps, typed argument records whose fields are dispatched as dynamic arguments to sub-commands (including through non-byte-aligned packed fields and validatable records), a dynamic sleep whose duration comes from the sequence argument, no-wait direct dispatch, a sequence that calls another sequence as a sub-sequence step, and sequences whose command response is statically deferred until completion. This example adheres to the schema shown in the previous section, and is commented to give clarification.

Note the optional top-level `with:` key, which lists additional Ada packages to “with” in the generated package. A step’s `arg:` expression is emitted verbatim into the generated Ada, so any package it references that is not pulled in automatically – such as an enum package needed to qualify an enumeration literal – must be listed here.

```

1  # Command Sequences Definition
2  # This file defines composite commands that execute sequences of sub-commands
3
4  name: Flight_Sequences
5  description: Common flight software command sequences
6
7  # Two frames: two sequences may run concurrently, and the generated summary
8  # packet type carries two frame summaries.
9  num_concurrent_sequences: 2
10
11 command_timeout_seconds: 10 # default wait for sub-command responses
12
13 sequences:
14   # Simple sequence with no arguments
15   - name: Sequence_A
16     description: Execute commands on component A
17     wait_for_command_completion: true
18     continue_on_failure: true
19
20     sequence:
21       - command: Component_A.Command_1
22       - command: Component_A.Command_2
23         arg: (Seconds => 3, Subseconds => 14)
24       - sleep_ms: 3000
25       - command: Component_A.Command_3
26         arg: (Value => 3)
27
28   - name: Sequence_B
29     description: Perform generic arguments with component A and B
30     wait_for_command_completion: true
31     continue_on_failure: false
32     arg_type: Sequence_B_Arg.T
33
34     sequence:
35       - command: Component_A.Command_3
36         arg: Arg.Component_A_Arg

```

```

37     - command: Component_B.Command_3
38       arg: Arg.Component_B_Arg
39
40 - name: Sequence_C
41   description: No-wait sequence for direct-dispatch testing
42   wait_for_command_completion: false
43   continue_on_failure: false
44
45   sequence:
46     - command: Component_A.Command_1
47     - command: Component_A.Command_1
48
49 - name: Sequence_D
50   description: Sequence with a dynamic sleep duration taken from the sequence
51   ↪ argument
52   wait_for_command_completion: true
53   continue_on_failure: false
54   arg_type: Packed_Natural.T
55
56   sequence:
57     - sleep_ms: Arg
58
59   # Calls another sequence. Each sequence is a command on the sequencer, so a
60   # sub-sequence step is a command step targeting the sequencer's own command.
61   # Sequence_C replies on start, so this sequence resumes as soon as it starts.
62 - name: Sequence_E
63   description: Call Sequence_C as a sub-sequence, then run a regular command
64   wait_for_command_completion: true
65   continue_on_failure: false
66
67   sequence:
68     - command: Sequencer.Sequence_C
69     - command: Component_A.Command_1
70
71   # Command response deferred until completion: the operator's reply carries
72   # the final success/failure instead of a start acknowledgement.
73 - name: Sequence_F
74   description: No-wait sequence with a statically deferred command response
75   wait_for_command_completion: false
76   continue_on_failure: false
77   response_behavior: send_after_sequence_completion
78
79   sequence:
80     - command: Component_A.Command_1
81
82   # Deferred response and waits on its sub-command, so the reply arrives when
83   # the sub-command completes (or the sequence aborts, times out, or is killed).
84 - name: Sequence_G
85   description: Waiting sequence with a statically deferred command response
86   wait_for_command_completion: true
87   continue_on_failure: false
88   response_behavior: send_after_sequence_completion
89
90   sequence:
91     - command: Component_A.Command_1
92
93   # Dynamic-arg repacking: the argument's leading 4-bit pad puts both leaf
94   # fields at non-byte-aligned offsets, so byte-offset slicing would corrupt
95   ↪ them.
96 - name: Sequence_H
97   description: Dynamic steps traversing misaligned 8-bit and 16-bit packed
98   ↪ fields

```

```

96     wait_for_command_completion: true
97     continue_on_failure: false
98     arg_type: Nibble_Test_Args.T
99
100    sequence:
101      - command: Component_A.Command_8
102        arg: Arg.Wide
103      - command: Component_A.Command_7
104        arg: Arg.Nibble
105
106    # First step is a static sleep, so wake-time overflow can be exercised
107    # without a command step's response deadline overflowing first.
108    - name: Sequence_I
109      description: A lone static sleep, for wake-time overflow coverage
110      wait_for_command_completion: true
111      continue_on_failure: false
112
113      sequence:
114        - sleep_ms: 3000
115
116    # The two sequences below take Validated_Args, whose Mode enumeration can
117    # fail validation, for the invalid-dynamic-argument failure paths.
118    - name: Sequence_J
119      description: Dynamic command argument resolved from a validatable record
120      wait_for_command_completion: true
121      continue_on_failure: false
122      arg_type: Validated_Args.T
123
124      sequence:
125        - command: Component_A.Command_3
126          arg: Arg.Duration
127
128    - name: Sequence_K
129      description: Dynamic sleep resolved from a validatable record
130      wait_for_command_completion: true
131      continue_on_failure: false
132      arg_type: Validated_Args.T
133
134      sequence:
135        - sleep_ms: Arg.Sleep_Duration

```

4 Example Output

The example input above produces the following Ada outputs. The package spec declares the `To_Arg` buffer-padding helper, the per-dynamic-step Resolver functions, a step array constant per sequence, the `Sequences_Table` constant (plus a stable `Sequences` access pointer), and the suite's `Config` constant, which bundles the table access with the suite's `num_concurrent_sequences`. The package body implements the Resolver functions: each validates the sequence's per-call argument buffer through the input type's generated `Validation.Valid`, deserializes it through `Serialization.From_Byte_Array`, and re-serializes the addressed field as the step's argument buffer (a sub-command's argument, or a `Packed_Natural` millisecond count for a dynamic sleep). Resolvers exist only for dynamic steps – a suite of purely static sequences declares none, and its body degenerates to `pragma No_Body;`.

The `Config` constant should be passed into the Simple Command Sequencer's `Init` procedure during assembly initialization, e.g. in the assembly YAML:

```

- type: Simple_Command_Sequencer
  name: My_Sequencer

```

```

init:
  - "Config => Test_Assembly_Command_Sequences_Example_Sequences.Config"

```

```

1  -- Auto-generated by Adamant command sequences generator.
2  -- Do not edit by hand.
3  -- Common flight software command sequences
4  with Simple_Sequencer_Types; use Simple_Sequencer_Types;
5  with Test_Assembly_Commands; use Test_Assembly_Commands;
6  with Command_Types;
7  with Basic_Types;
8  with Sequence_Enums;
9  with Ada.Real_Time;
10 with Sys_Time_32;
11 with Packed_U32;
12 with Sequence_B_Arg;
13 with Packed_Natural;
14 with Nibble_Test_Args;
15 with Packed_U16;
16 with Packed_Nibble;
17 with Validated_Args;
18 package Test_Assembly_Command_Sequences_Example_Sequences is
19   -- Make the "&" operator on Byte_Array visible for To_Arg below.
20   use type Basic_Types.Byte_Array;
21
22   -----
23   -- Pad a serialized argument out to a full command argument buffer. An
24   -- expression function so the step arrays below can call it at elaboration.
25   -----
26   function To_Arg (Bytes : Basic_Types.Byte_Array) return
27   ↪ Command_Types.Command_Arg_Buffer_Type is
28   ↪ (Command_Types.Command_Arg_Buffer_Type (Bytes & Basic_Types.Byte_Array'(1
29   ↪   .. Command_Types.Command_Arg_Buffer_Type'Length - Bytes'Length =>
30   ↪   0)));
31   -----
32   -- Resolvers - one per dynamic step, each extracting one field of the
33   -- sequence argument.
34   -----
35   function Sequence_B_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
36   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
37   function Sequence_B_Step_1_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
38   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
39   function Sequence_D_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
40   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
41   function Sequence_H_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
42   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
43   function Sequence_H_Step_1_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
44   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
45   function Sequence_J_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
46   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
47   function Sequence_K_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
48   ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean;
49   -----
50   -- Step arrays - one constant per sequence.
51   -----
52   Sequence_A_Steps : aliased constant Step_Array :=
53   [
54     0 =>
55       (Kind      => Command_Step,
56        Id        => Component_A_Command_1,
57        Arg_Length => 0,
58        Arg       => [others => 0]),
59     1 =>

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```

50         (Kind      => Command_Step,
51          Id        => Component_A_Command_2,
52          Arg_Length => Sys_Time_32.Serialization.Serialized_Length,
53          Arg        => To_Arg (Sys_Time_32.Serialization.To_Byte_Array
54                               ↳ (((Seconds => 3, Subseconds => 14))))),
55     2 =>
56         (Kind      => Sleep,
57          Id        => 0,
58          Arg_Length => 0,
59          Sleep_Arg  => 3000),
60     3 =>
61         (Kind      => Command_Step,
62          Id        => Component_A_Command_3,
63          Arg_Length => Packed_U32.Serialization.Serialized_Length,
64          Arg        => To_Arg (Packed_U32.Serialization.To_Byte_Array (((Value
65                               ↳ => 3))))))
66 ];
67 Sequence_B_Steps : aliased constant Step_Array :=
68 [
69     0 =>
70         (Kind      => Runtime_Argument_Command_Step,
71          Id        => Component_A_Command_3,
72          Arg_Length => Packed_U32.Serialization.Serialized_Length,
73          Resolver  => Sequence_B_Step_0_Resolver_T'Access),
74     1 =>
75         (Kind      => Runtime_Argument_Command_Step,
76          Id        => Component_B_Command_3,
77          Arg_Length => Packed_U32.Serialization.Serialized_Length,
78          Resolver  => Sequence_B_Step_1_Resolver_T'Access)
79 ];
80 Sequence_C_Steps : aliased constant Step_Array :=
81 [
82     0 =>
83         (Kind      => Command_Step,
84          Id        => Component_A_Command_1,
85          Arg_Length => 0,
86          Arg        => [others => 0]),
87     1 =>
88         (Kind      => Command_Step,
89          Id        => Component_A_Command_1,
90          Arg_Length => 0,
91          Arg        => [others => 0])
92 ];
93 Sequence_D_Steps : aliased constant Step_Array :=
94 [
95     0 =>
96         (Kind      => Runtime_Sleep,
97          Id        => 0,
98          Arg_Length => Packed_Natural.Serialization.Serialized_Length,
99          Sleep_Resolver => Sequence_D_Step_0_Resolver_T'Access)
100 ];
101 Sequence_E_Steps : aliased constant Step_Array :=
102 [
103     0 =>
104         (Kind      => Command_Step,
105          Id        => Sequencer_Sequence_C,
106          Arg_Length => 0,
107          Arg        => [others => 0]),
108     1 =>
109         (Kind      => Command_Step,
110          Id        => Component_A_Command_1,

```

```

109     Arg_Length => 0,
110     Arg         => [others => 0])
111 ];
112 Sequence_F_Steps : aliased constant Step_Array :=
113 [
114     0 =>
115         (Kind          => Command_Step,
116          Id            => Component_A_Command_1,
117          Arg_Length    => 0,
118          Arg           => [others => 0])
119 ];
120 Sequence_G_Steps : aliased constant Step_Array :=
121 [
122     0 =>
123         (Kind          => Command_Step,
124          Id            => Component_A_Command_1,
125          Arg_Length    => 0,
126          Arg           => [others => 0])
127 ];
128 Sequence_H_Steps : aliased constant Step_Array :=
129 [
130     0 =>
131         (Kind          => Runtime_Argument_Command_Step,
132          Id            => Component_A_Command_8,
133          Arg_Length    => Packed_U16.Serialization.Serialized_Length,
134          Resolver      => Sequence_H_Step_0_Resolver_T'Access),
135     1 =>
136         (Kind          => Runtime_Argument_Command_Step,
137          Id            => Component_A_Command_7,
138          Arg_Length    => Packed_Nibble.Serialization.Serialized_Length,
139          Resolver      => Sequence_H_Step_1_Resolver_T'Access)
140 ];
141 Sequence_I_Steps : aliased constant Step_Array :=
142 [
143     0 =>
144         (Kind          => Sleep,
145          Id            => 0,
146          Arg_Length    => 0,
147          Sleep_Arg     => 3000)
148 ];
149 Sequence_J_Steps : aliased constant Step_Array :=
150 [
151     0 =>
152         (Kind          => Runtime_Argument_Command_Step,
153          Id            => Component_A_Command_3,
154          Arg_Length    => Packed_U32.Serialization.Serialized_Length,
155          Resolver      => Sequence_J_Step_0_Resolver_T'Access)
156 ];
157 Sequence_K_Steps : aliased constant Step_Array :=
158 [
159     0 =>
160         (Kind          => Runtime_Sleep,
161          Id            => 0,
162          Arg_Length    => Packed_Natural.Serialization.Serialized_Length,
163          Sleep_Resolver => Sequence_K_Step_0_Resolver_T'Access)
164 ];
165 -----
166 -- Sequences_Table
167 -----
168 Sequences_Table : aliased constant Sequences_Type :=
169 [

```

```

170     0 =>
171         (Wait_For_Cmd_Resp    => True,
172          Abort_On_Failed_Cmd => False,
173          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
174          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
175          Arg_Length           => 0,
176          Steps                => Sequence_A_Steps'Access),
177     1 =>
178         (Wait_For_Cmd_Resp    => True,
179          Abort_On_Failed_Cmd => True,
180          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
181          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
182          Arg_Length           => Sequence_B_Arg.Serialization.Serialized_Length,
183          Steps                => Sequence_B_Steps'Access),
184     2 =>
185         (Wait_For_Cmd_Resp    => False,
186          Abort_On_Failed_Cmd => True,
187          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
188          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
189          Arg_Length           => 0,
190          Steps                => Sequence_C_Steps'Access),
191     3 =>
192         (Wait_For_Cmd_Resp    => True,
193          Abort_On_Failed_Cmd => True,
194          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
195          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
196          Arg_Length           => Packed_Natural.Serialization.Serialized_Length,
197          Steps                => Sequence_D_Steps'Access),
198     4 =>
199         (Wait_For_Cmd_Resp    => True,
200          Abort_On_Failed_Cmd => True,
201          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
202          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
203          Arg_Length           => 0,
204          Steps                => Sequence_E_Steps'Access),
205     5 =>
206         (Wait_For_Cmd_Resp    => False,
207          Abort_On_Failed_Cmd => True,
208          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
209          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
210          Arg_Length           => 0,
211          Steps                => Sequence_F_Steps'Access),
212     6 =>
213         (Wait_For_Cmd_Resp    => True,
214          Abort_On_Failed_Cmd => True,
215          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
216          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
217          Arg_Length           => 0,
218          Steps                => Sequence_G_Steps'Access),
219     7 =>
220         (Wait_For_Cmd_Resp    => True,
221          Abort_On_Failed_Cmd => True,
222          Command_Timeout     => Ada.Real_Time.Milliseconds (10000),
223          Response_Behavior    => Sequence_Enums.Sequence_Response_Behavior.Send,
224          Arg_Length           => 0,
225          Steps                => Sequence_H_Steps'Access),

```

```

224     Arg_Length      =>
225     ↪ Nibble_Test_Args.Serialization.Serialized_Length,
226     Steps           => Sequence_H_Steps'Access),
227 8 =>
228     (Wait_For_Cmd_Resp => True,
229     Abort_On_Failed_Cmd => True,
230     Command_Timeout    => Ada.Real_Time.Milliseconds (10000),
231     Response_Behavior  => Sequence_Enums.Sequence_Response_Behavior.Send,
232     ↪ _After_Sequence_Start,
233     Arg_Length         => 0,
234     Steps              => Sequence_I_Steps'Access),
235 9 =>
236     (Wait_For_Cmd_Resp => True,
237     Abort_On_Failed_Cmd => True,
238     Command_Timeout    => Ada.Real_Time.Milliseconds (10000),
239     Response_Behavior  => Sequence_Enums.Sequence_Response_Behavior.Send,
240     ↪ _After_Sequence_Start,
241     Arg_Length         => Validated_Args.Serialization.Serialized_Length,
242     Steps              => Sequence_J_Steps'Access),
243 10 =>
244     (Wait_For_Cmd_Resp => True,
245     Abort_On_Failed_Cmd => True,
246     Command_Timeout    => Ada.Real_Time.Milliseconds (10000),
247     Response_Behavior  => Sequence_Enums.Sequence_Response_Behavior.Send,
248     ↪ _After_Sequence_Start,
249     Arg_Length         => Validated_Args.Serialization.Serialized_Length,
250     Steps              => Sequence_K_Steps'Access)
251 ];
252
253 Sequences : constant Sequences_Access := Sequences_Table'Access;
254
255 -- Passed to Simple_Command_Sequencer.Init. num_concurrent_sequences also
256 -- sizes the generated
257 ↪ Test_Assembly_Command_Sequences_Example_Sequences_Summary_Record packet
258 ↪ type, so the frame
259 -- pool and the packet layout cannot disagree.
260 Config : constant Sequencer_Config :=
261     (Sequences => Sequences_Table'Access,
262     Num_Concurrent_Sequences => 2);
263 end Test_Assembly_Command_Sequences_Example_Sequences;

```

```

1  -- Auto-generated by Adamant command sequences generator.
2  -- Do not edit by hand.
3  with Interfaces;
4  with Sequence_B_Arg.Validation;
5  with Packed_Natural.Validation;
6  with Nibble_Test_Args.Validation;
7  with Validated_Args.Validation;
8  package body Test_Assembly_Command_Sequences_Example_Sequences is
9
10     -----
11     -- Resolvers - one per dynamic step. Each validates and deserializes the
12     -- sequence argument, then returns the addressed field serialized (a
13     -- sub-command argument, or a Packed_Natural millisecond count for a sleep).
14     -----
15     function Sequence_B_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
16     ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
17     Errant_Field : Interfaces.Unsigned_32;
18     Input : Sequence_B_Arg.T;
19     Valid : Boolean := Sequence_B_Arg.Validation.Valid (Bytes (Bytes'First ..
20     ↪ Bytes'First + Sequence_B_Arg.Serialization.Serialized_Length - 1),
21     ↪ Errant_Field);

```

```

19   begin
20       if Valid then
21           Input := Sequence_B_Arg.Serialization.From_Byte_Array (Bytes
22               ↪ (Bytes'First .. Bytes'First +
23               ↪ Sequence_B_Arg.Serialization.Serialized_Length - 1));
24           Args := To_Arg (Packed_U32.Serialization.To_Byte_Array
25               ↪ (Input.Component_A_Arg));
26       end if;
27       return Valid;
28   end Sequence_B_Step_0_Resolver_T;
29   function Sequence_B_Step_1_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
30       ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
31       Errant_Field : Interfaces.Unsigned_32;
32       Input : Sequence_B_Arg.T;
33       Valid : Boolean := Sequence_B_Arg.Validation.Valid (Bytes (Bytes'First ..
34           ↪ Bytes'First + Sequence_B_Arg.Serialization.Serialized_Length - 1),
35           ↪ Errant_Field);
36   begin
37       if Valid then
38           Input := Sequence_B_Arg.Serialization.From_Byte_Array (Bytes
39               ↪ (Bytes'First .. Bytes'First +
40               ↪ Sequence_B_Arg.Serialization.Serialized_Length - 1));
41           Args := To_Arg (Packed_U32.Serialization.To_Byte_Array
42               ↪ (Input.Component_B_Arg));
43       end if;
44       return Valid;
45   end Sequence_B_Step_1_Resolver_T;
46   function Sequence_D_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
47       ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
48       Errant_Field : Interfaces.Unsigned_32;
49       Input : Packed_Natural.T;
50       Valid : Boolean := Packed_Natural.Validation.Valid (Bytes (Bytes'First ..
51           ↪ Bytes'First + Packed_Natural.Serialization.Serialized_Length - 1),
52           ↪ Errant_Field);
53   begin
54       if Valid then
55           Input := Packed_Natural.Serialization.From_Byte_Array (Bytes
56               ↪ (Bytes'First .. Bytes'First +
57               ↪ Packed_Natural.Serialization.Serialized_Length - 1));
58           Args := To_Arg (Packed_Natural.Serialization.To_Byte_Array (Input));
59       end if;
60       return Valid;
61   end Sequence_D_Step_0_Resolver_T;
62   function Sequence_H_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
63       ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
64       Errant_Field : Interfaces.Unsigned_32;
65       Input : Nibble_Test_Args.T;
66       Valid : Boolean := Nibble_Test_Args.Validation.Valid (Bytes (Bytes'First
67           ↪ .. Bytes'First + Nibble_Test_Args.Serialization.Serialized_Length -
68           ↪ 1), Errant_Field);
69   begin
70       if Valid then
71           Input := Nibble_Test_Args.Serialization.From_Byte_Array (Bytes
72               ↪ (Bytes'First .. Bytes'First +
73               ↪ Nibble_Test_Args.Serialization.Serialized_Length - 1));
74           Args := To_Arg (Packed_U16.Serialization.To_Byte_Array (Input.Wide));
75       end if;
76       return Valid;
77   end Sequence_H_Step_0_Resolver_T;
78   function Sequence_H_Step_1_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
79       ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is

```

```

60     Errant_Field : Interfaces.Unsigned_32;
61     Input : Nibble_Test_Args.T;
62     Valid : Boolean := Nibble_Test_Args.Validation.Valid (Bytes (Bytes'First
        ↪ .. Bytes'First + Nibble_Test_Args.Serialization.Serialized_Length -
        ↪ 1), Errant_Field);
63 begin
64     if Valid then
65         Input := Nibble_Test_Args.Serialization.From_Byte_Array (Bytes
            ↪ (Bytes'First .. Bytes'First +
            ↪ Nibble_Test_Args.Serialization.Serialized_Length - 1));
66         Args := To_Arg (Packed_Nibble.Serialization.To_Byte_Array
            ↪ (Input.Nibble));
67     end if;
68     return Valid;
69 end Sequence_H_Step_1_Resolver_T;
70 function Sequence_J_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
    ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
71     Errant_Field : Interfaces.Unsigned_32;
72     Input : Validated_Args.T;
73     Valid : Boolean := Validated_Args.Validation.Valid (Bytes (Bytes'First ..
        ↪ Bytes'First + Validated_Args.Serialization.Serialized_Length - 1),
        ↪ Errant_Field);
74 begin
75     if Valid then
76         Input := Validated_Args.Serialization.From_Byte_Array (Bytes
            ↪ (Bytes'First .. Bytes'First +
            ↪ Validated_Args.Serialization.Serialized_Length - 1));
77         Args := To_Arg (Packed_U32.Serialization.To_Byte_Array
            ↪ (Input.Duration));
78     end if;
79     return Valid;
80 end Sequence_J_Step_0_Resolver_T;
81 function Sequence_K_Step_0_Resolver_T (Bytes : Basic_Types.Byte_Array; Args :
    ↪ out Command_Types.Command_Arg_Buffer_Type) return Boolean is
82     Errant_Field : Interfaces.Unsigned_32;
83     Input : Validated_Args.T;
84     Valid : Boolean := Validated_Args.Validation.Valid (Bytes (Bytes'First ..
        ↪ Bytes'First + Validated_Args.Serialization.Serialized_Length - 1),
        ↪ Errant_Field);
85 begin
86     if Valid then
87         Input := Validated_Args.Serialization.From_Byte_Array (Bytes
            ↪ (Bytes'First .. Bytes'First +
            ↪ Validated_Args.Serialization.Serialized_Length - 1));
88         Args := To_Arg (Packed_Natural.Serialization.To_Byte_Array
            ↪ (Input.Sleep_Duration));
89     end if;
90     return Valid;
91 end Sequence_K_Step_0_Resolver_T;
92 end Test_Assembly_Command-Sequences_Example-Sequences;
93

```

5 Per-Sequence First-Class Commands

In addition to the sequence tables above, this same model drives the autocoded synthesis of one first-class command per declared sequence on the Simple Command Sequencer instance. Discovery is build-path-driven: the sequencer's commands model queries the model database for every `*.command_sequences.yaml` in scope, so a naked component build (no assembly) sees no sequences and produces only the four static commands (Run_Sequence, Kill_All_Sequences,

Kill_Frame, Set_Summary_Packet_Period), while an assembly build whose path includes a sequences suite adds one command per sequence. Each per-sequence command (e.g. Sequence_B) takes the sequence's own arg_type verbatim as its command argument (or no argument for an argless sequence) – a user-written, normally-registered type, so no argument records are generated or registered anywhere in the pipeline. At run time the component copies the received argument buffer unchanged into a generic Run_Sequence_Arg.T and dispatches through the existing Run_Sequence backbone using the sequence's runtime slot index. Run_Sequence rejects any argument whose length differs from the serialized length of the sequence's arg_type (recorded in the autocoded table) with an Invalid_Sequence_Argument_Length event; the response behavior comes from the sequence's static configuration in the same table (see Sequence Response Behavior below). A companion generated package, <suite>_Commands, carries the operator-side builder surface (per-sequence command id getters and Command.T constructors) for unit tests and other on-board callers – split from the sequences package because the sequencer component itself never uses the builders. No per-sequence Ada needs to be hand-written:

```

1  -- Auto-generated by Adamant command sequences generator.
2  -- Do not edit by hand.
3  -- Common flight software command sequences
4  --
5  -- Operator-side builders (id getters and Command.T constructors) for the
6  -- Test_Assembly_Command_Sequences_Example_Sequences suite's per-sequence
   ↪ commands, for unit tests and on-board
7  -- callers. The sequencer component itself does not use this package.
8  with Command;
9  with Command_Types;
10 with Nibble_Test_Args;
11 with Packed_Natural;
12 with Sequence_B_Arg;
13 with Validated_Args;
14 package Test_Assembly_Command_Sequences_Example_Sequences_Commands is
15
16     -- Set Id_Base to the sequencer instance's command id base. Each sequence
17     -- command id is Id_Base + Num_Commands + declaration index.
18     type Instance is tagged private;
19
20     -- Set the sequencer instance's command id base so constructed commands
   ↪ carry
21     -- ids that match what the component registered at runtime.
22     procedure Set_Id_Base (Self : in out Instance; Id_Base : in
   ↪ Command_Types.Command_Id);
23
24     -- Set the operator source id stamped into the header of constructed commands
25     -- (defaults to 0). Mirrors the command suite's Set_Source_Id.
26     procedure Set_Source_Id (Self : in out Instance; Source_Id : in
   ↪ Command_Types.Command_Source_Id);
27
28     function Get_Sequence_A_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
29     function Sequence_A (Self : in Instance) return Command.T;
30     function Get_Sequence_B_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
31     function Sequence_B (Self : in Instance; Arg : in Sequence_B_Arg.T) return
   ↪ Command.T;
32     function Get_Sequence_C_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
33     function Sequence_C (Self : in Instance) return Command.T;
34     function Get_Sequence_D_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
35     function Sequence_D (Self : in Instance; Arg : in Packed_Natural.T) return
   ↪ Command.T;
36     function Get_Sequence_E_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;

```

```

37 function Sequence_E (Self : in Instance) return Command.T;
38 function Get_Sequence_F_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
39 function Sequence_F (Self : in Instance) return Command.T;
40 function Get_Sequence_G_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
41 function Sequence_G (Self : in Instance) return Command.T;
42 function Get_Sequence_H_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
43 function Sequence_H (Self : in Instance; Arg : in Nibble_Test_Args.T) return
   ↪ Command.T;
44 function Get_Sequence_I_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
45 function Sequence_I (Self : in Instance) return Command.T;
46 function Get_Sequence_J_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
47 function Sequence_J (Self : in Instance; Arg : in Validated_Args.T) return
   ↪ Command.T;
48 function Get_Sequence_K_Id (Self : in Instance) return
   ↪ Command_Types.Command_Id;
49 function Sequence_K (Self : in Instance; Arg : in Validated_Args.T) return
   ↪ Command.T;
50
51 private
52   type Instance is tagged record
53     Id_Base : Command_Types.Command_Id := 0;
54     Source_Id : Command_Types.Command_Source_Id := 0;
55   end record;
56 end Test_Assembly_Command_Sequences_Example_Sequences_Commands;

```

6 Sequences Calling Sequences

Because every sequence is a first-class command on the sequencer instance, a sequence step may invoke another sequence simply by targeting the sequencer's own synthesized command, e.g. `command: My_Sequencer.Sequence_C`. No special syntax or generator support is involved – the step resolves like any other `Component.Command` reference, provided the sequencer instance is declared in the assembly. At run time the sub-sequence command is emitted out the sequencer's `Command_T_Send` connector, routed through the command router, and arrives back at the sequencer's `Command_T_Recv_Async`, where it claims its own frame and executes concurrently with (and independently of) the calling sequence. If the calling sequence waits for command completion, the sub-sequence's command response (per its static `response_behavior` configuration, see below) is what wakes it. Nesting depth is naturally capped by `Num_Concurrent_Sequences`: a sub-sequence call when no frame is available is rejected with a `No_Frame_Available` event and a `Failure` response to the calling frame.

7 Sequence Response Behavior

When the sequencer replies to the command that started a sequence is controlled by the `Sequence_Response_Behavior` enumeration. It is static per-sequence configuration: the optional `response_behavior` field on each sequence in the YAML, baked into the autocoded sequence table and used by both the synthesized per-sequence commands and the generic `Run_Sequence` command:

- `Send_After_Sequence_Start` (the conventional default) – the command response is sent immediately once the sequence has been accepted and started on a frame. This is the right choice for fire-and-forget operation and for sub-sequence calls where the caller should proceed as soon as the sub-sequence is running.

- `Send_After_Sequence_Completion` – the command response is withheld until the sequence ends, and its status reflects the outcome: `Success` when the final step completes, or `Failure` when the sequence is aborted by a failed sub-command, times out waiting for a sub-command response, fails to resolve or validate a dynamic argument, encounters an out-of-range sleep or timeout, or is halted by `Kill_All_Sequences` or `Kill_Frame`. The sequencer captures the originating command’s source and id per frame, so concurrent deferred sequences each reply to their own operator.

A sequence step may forward a field of the parent sequence’s argument to the sub-command it dispatches, using the `Arg.field.subfield` syntax in the step’s `arg:.` Each such reference produces a typed Resolver in the generated package: at run time the sequencer dispatches that Resolver, which first validates the per-call argument buffer through the input type’s generated `Validation.Valid` (a failure aborts the sequence with an `Invalid_Dynamic_Command_Argument` event), then deserializes it through the input type’s safe `Serialization.From_Byte_Array`, navigates to the addressed field, and re-serializes it as the sub-command’s argument buffer. The resolver’s signature accepts only a `Basic_Types.Byte_Array` – there is no ‘Address reinterpretation’ – so the dynamic-argument machinery is type-safe by construction; a resolver cannot overlay an arbitrary record onto raw memory, only deserialize a byte array through the input type’s generated deserializer.

8 Sleep Steps

A sequence step may be a sleep instead of a command: `sleep_ms:` pauses the executing frame without issuing any bus traffic. Sleep is not a bus command; it is interpreted intrinsically by the sequencer frame state machine. The duration takes two forms. A *static* sleep (`sleep_ms: 3000`) is a plain integer number of milliseconds fixed at build time; the model bounds it to $0 \dots 2^{31} - 1$ so it always fits an `Ada.Real_Time.Time_Span` by construction, leaving wake-time overflow of `Sys_Time` arithmetic as its only run-time failure. A *dynamic* sleep (`sleep_ms: Arg.Settle_Time`) instead references the sequence’s per-call argument with the same `Arg[.field]` syntax as dynamic command arguments; the referenced field must be a `Packed_Natural.T`, resolved at execution time through the same validating Resolver machinery, so the duration is range-checked as part of argument validation (a failure aborts the sequence with an `Invalid_Dynamic_Sleep_Argument` event) and always fits a `Time_Span`. A sleep whose wake-time computation overflows ends the sequence with a `Sequence_Out_Of_Range_Sleep` event and a `Failure` status to any pending deferred reply – continuing would leave the frame parked on a stale wake time.

9 Summary Packet Type

The sequencer emits a periodic `Summary_Packet` containing one `Sequence_Frame_Summary` record per sequence frame, in frame order. Its layout therefore depends on the suite’s `num_concurrent_sequences`, so the generator also autocodes the packet’s ground/documentation record type, `<suite>_summary_record.record.yaml`, alongside the suite’s other outputs. The flight software never imports this type – the component fills the packet buffer directly from its frame array – but ground tools and assembly documentation need it. At assembly load time the sequencer’s packets model resolves an instance’s `Summary_Packet` type to the summary record of the suite named by the instance’s `Config init` parameter. Because the frame pool and the packet layout both come from the suite model’s single `num_concurrent_sequences` value, they can never disagree; the model also rejects at build time any suite whose frame summaries would not fit in one packet buffer (the same bound the `Num_Concurrent_Sequences_Type` subtype enforces at `Init`). The example input above generates the following record model:

```

1  ---
2  description: Autocoded summary packet type for the
   ↳ Test_Assembly_Command_Sequences_Example_Sequences command sequences suite.
   ↳ Contains one Sequence_Frame_Summary record per sequence frame (2 total, from
   ↳ the suite's num_concurrent_sequences), in frame order.
3  fields:
```

```
4 - name: Frame_0_Summary
5   description: "A summary of the state of sequence frame 0."
6   type: Sequence_Frame_Summary.T
7 - name: Frame_1_Summary
8   description: "A summary of the state of sequence frame 1."
9   type: Sequence_Frame_Summary.T
10
```